XIE LIDONG

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Personal

Date of Birth: 6 October 2002.
Place of Birth: Sichuan Province, China.
Citizenship:China.

Education

2026-Present Master in Mathematics (Financial Mathematics track),
University of Luxembourg, Luxembourg.

2020-2024 Bachelor of Science in Data Science and Big Data Technology, Ningbo University of Technology, China.

English Level:

IELTS Overall Band Score 7.0 |C1 Level (PDF-6)

University score:

Cumulative GPA:3.07

Cumulative Average Score:80.25/100

Math and Distinguished Courses in university:

Math:
Nonparametric Statistics : 92/100
Advanced Mathematics AI: 90/100
Regression and Multivariate Statistics Lab: 85/100
Data Structures and Algorithms:85/100
Time Series Analysis : 84/100
Mathematical Modeling practice : 81/100
Advanced Mathematics AII：81/100
Linear AlgebraI：80/100
Statistical Models :79/100
Mathematical Modeling ： 78/100
Probability Theory and Mathematical Statistics :78/100
Regression and Multivariate Statistics : 77/100
Numerical Analysis:77/100
Time Series Analysis Lab:77/100
Applied Bayesian Statistics :73/100

Other Distinguished Courses:
C Language Programming :95/100
Business English :93/100
College English Ⅱ:90/100
Multinational Corporate Finance :90/100
Electronic Craft Practice:90/100
Circuit Experiment :86/100
Electronic CAD: 85/100
Hands-On Big Data Mining :85/100
College English Ⅰ:83/100
Hands-On Machine Learning:83/100

Awards & Honors:

2025 Completed Harvard University online course PH125.3x: Data Science: Probability with 100/100 | Earned Certificate. (PDF-7)

2025 Completed LSE online course MT105b: Mathematics 1 Part 2: Integral calculus, algebra, and applications with a score of 98/100 | Earned Certificate . (PDF-9)

2024 Received a conditional offer from UNSW for the Master of Artificial Intelligence program. (PDF-11)

2020 Freshman Scholarship (¥5000), Ningbo University of Technology| Awarded for exceeding the first-tier college admission score line. (PDF-12)

2024 First-Class Scholarship for Postgraduate Admission (¥2000), Ningbo University of Technology | Awarded for obtaining offer from QS top 50 university —— UNWS. (PDF-13)

2020 Outstanding Cadre of the University Student Security Committee in October, Ningbo University of Technology | Excellent work performance (PDF-15)

2020 Outstanding Cadre of the University Student Security Committee in November, Ningbo University of Technology | Excellent work performance (PDF-14)

2022 Vice Minister of the University Student Security Committee, Ningbo University of Technology | Promote

Personal & Technical Projects:

1 **Crypto Volatility & VaR Analysis Dashboard**

Python, Streamlit, Time Series Analysis, GARCH(1,1) Model, Volatility Modeling, Value-at-Risk (VaR), Rolling Window Forecast, MAE Evaluation, Fat-Tailed t-Distribution, Yahoo Finance , Data Visualization, Cryptocurrency Risk Analytics, Dynamic Risk Assessment

This project develops a volatility modeling and extreme risk quantification system for the cryptocurrency market. Using the GARCH(1,1) model to capture volatility clustering, it computes dynamic Value-at-Risk (VaR) and forecasts short-term

volatility, delivering interpretable, validated quantitative risk analytics tools for highly volatile assets.
Built a full-cycle interactive web analytics dashboard with Streamlit for Bitcoin and Ethereum, supporting customizable date-range data fetching and preprocessing from Yahoo Finance. Implemented the GARCH(1,1) time series model to fit volatility dynamics, estimating key parameters and long-run unconditional volatility to quantify market risk persistence.
Designed dual VaR computation frameworks under normal and fat-tailed t-distributions to quantify extreme loss potential, with backtesting via VaR violation rate analysis.
Developed a rolling-window forecasting pipeline for short-term volatility prediction, and applied MAE to objectively measure and validate prediction accuracy.
Created multi-panel comparative visualizations to contrast fitted GARCH volatility, raw rolling volatility, and out-of-sample predictions, with parameter annotations for enhanced financial interpretability.

Streamlit:

https://cryptocurrency-volatility-forecasting-dynamic-var-risk-managem.streamlit.app/

Github:

https://github.com/XIE-LIDONG/Cryptocurrency-Volatility-Forecasting-Dynamic-VaR-Risk-Management

2 Option Pricing Analytics Engine & Interactive Dashboard

Python, Streamlit, NumPy, Pandas, Financial Engineering, Black-Scholes Model, Monte Carlo Simulation, Binomial Tree, Trinomial Tree, Finite Difference Method (FDM), Longstaff-Schwartz Method (LSM), European/American Options, Early Exercise Optimization, Convergence Analysis

Built an option pricing engine supporting European/American options with 7 core algorithms (Black-Scholes, Monte Carlo, Binomial Tree, Trinomial Tree, Finite Difference (European/American), Longstaff-Schwartz Method (LSM)).

Created an interactive Streamlit dashboard with real-time parameter adjustment, dynamic UI, and side-by-side comparison of different pricing methodologies. Implemented American option pricing with early exercise optimization using LSM Monte Carlo and Finite Difference Methods.

Added practical features: progress tracking for large-scale simulations, 95% confidence intervals, convergence analysis against analytical benchmarks, and educational use cases.

Designed clear result visualizations with parameter summary cards and comparative performance metrics (price accuracy, computation time).

Streamlit:

https://option-pricing-tdkclhc4zgvbkpjdwepb47.streamlit.app/

Github:https://github.com/XIE-LIDONG/Option-Pricing

3 Real-Time Mask Detection System with YOLO11

Python, Streamlit, Ultralytics YOLO11, OpenCV, NumPy, Deep Learning, Computer Vision

Based on the public Face-Mask-Detection dataset from Kaggle, a high-precision mask detection model was custom fine-tuned using the YOLO11 deep learning framework. The model achieves high-accuracy recognition for three core detection scenarios: 96.9% accuracy for correctly worn masks (with_mask), 89.2% accuracy for improperly worn masks (mask_weared_incorrect), and 93.4% accuracy for no masks (without_mask).

In terms of model engineering implementation, the training phase adopted a unified 640 × 640 image input size with 30 (adjustable through code) training epochs. Leveraging GPU acceleration on the Kaggle platform significantly improved model iteration efficiency. The trained model was packaged as a Mask_detector.pt file (40.5 MB in size) and deployed locally via the Streamlit framework to build a real-time mask detection system supporting video stream monitoring.

The system implements camera access and video stream processing based on OpenCV (compatible with Windows camera drivers), and completes efficient image data calculation with NumPy. It achieves low-latency real-time inference even in a CPU-only hardware environment. Meanwhile, it supports dynamic adjustment of detection confidence thresholds and provides interactive operation controls (e.g., Start/Stop Detection) through the Streamlit visual interface, enabling end-to-end full-process real-time monitoring of mask-wearing status in real-world scenarios.

Business Value: Replaces manual inspection with an overall real-time detection accuracy of 95%, improves safety management efficiency, and effectively reduces related labor costs.

Github:https://github.com/XIE-LIDONG/Mask-Detection

4 Google Maps Data Crawler

Python, Selenium, Web Scraping, Google Maps Data Extraction, Anti-Bot Detection, Headless Browser, Regex,Data Deduplication

Developed a Streamlit-based Google Maps data scraping tool that automatically extracts business information (name, address, phone number) based on location and keyword searches.

Implemented browser automation using Selenium with anti-detection measures and headless mode support to enhance scraping efficiency and stealth.

Created intelligent scrolling and deduplication mechanisms to ensure the completeness and uniqueness of collected data.

Designed a phone number extraction logic based on digit length (≥7 digits), with priority given to numbers in international format.

Built real-time progress display and Excel export functionality to enable immediate data usability.

Provided configurable stop conditions to optimize crawling efficiency and reduce unnecessary resource consumption.

Enabled automated data collection to support market research, competitive analysis, and lead generation scenarios.

Github:https://github.com/XIE-LIDONG/Googlemaps-searching-tool

5 Word Document Translation Tool

Python, Streamlit, Google Translate API, Multi-threading, python-docx, Batch Processing

A user-friendly Streamlit-based web application designed for translating DOCX documents while preserving the original formatting. The tool supports 8 major languages (Chinese, English, French, German, Spanish, Arabic, Japanese, Korean) and addresses traditional translation pain points such as format loss and inefficient manual processing.

Format-Preserving Translation:Parses DOCX documents to extract text from paragraphs and tables, then reconstructs the translated content back into the original document structure without formatting loss.Provides clear guidance for PDF users to convert files to DOCX via ILovePDF.

Configurable Performance Controls:Allows users to adjust thread count (1-3 threads) and batch size (20-100 text segments per batch) to balance translation speed and system stability.Implements multi-threaded batch translation using ThreadPoolExecutor for efficient processing of large documents.

Real-Time User Experience:Displays real-time translation progress (completed/total segments) and performance metrics (total time, translation speed in segments/second).Intuitive UI with file upload, language selection dropdowns, and one-click translation/download buttons.Automatic temporary file cleanup to avoid storage bloat and ensure system hygiene.

User-Centric Design:Responsive layout with file size display, translation direction labeling, and clear success/error notifications.Customized output filename (includes source-target language pair) for easy file management.

Business Value:Ideal for translating multilingual technical documents, business contracts, and official documents. Eliminates manual copy-paste translation work, improves translation efficiency by 60%+, and ensures document formatting integrity, reducing post-translation editing costs.

Streamlit APP:

https://docx-translator-app-id89euijvysspmvapptarhe.streamlit.app/

Github:https://github.com/XIE-LIDONG/docx-translator-app

6 Truck Purchase Intention Collection Platform

Python, Streamlit, Feishu API, Bilingual Web Application ,Bilingual UI (Arabic/English)

Developed a bilingual (Arabic/English) truck inquiry application using Streamlit, which automatically provides corresponding technical specification PDFs to customers after they submit contact information and vehicle preferences.

Integrated Feishu Webhook for real-time sales notifications, automatically pushing customer inquiry data (name, phone number, selected truck models) to sales teams.

Designed a structured vehicle-PDF mapping database to enable intelligent document matching.Implemented Saudi Arabia timezone timestamp processing to ensure accurate notification timing.

Frontend: Streamlit with customized CSS for bilingual interfaces, including support for Arabic right-to-left (RTL) layout.

Backend: Python-based processing of form data, PDF file matching, and Webhook API calls.

Integration: Feishu API for team collaboration notifications and local file system management for technical document storage and delivery.

Business Value: Digitized traditional vehicle sales inquiry processes, delivering instant technical documentation to customers and high-quality real-time leads to sales teams, thereby improving customer acquisition efficiency in Middle Eastern markets.

Streamlit:

https://purchase-intention-nq6c2k2pt5rn9yvv8tel6t.streamlit.app/

Github:

https://github.com/XIE-LIDONG/truck-customer-information-collect

7 Truck Sales Data Analytics Dashboard

Python, Streamlit, Pandas, Matplotlib,Data Visualization, KPI Tracking, Time Series Analysis

Developed a truck sales data analytics dashboard for commercial truck sales management, enabling panoramic visualization, multi-dimensional in-depth analysis, and real-time KPI monitoring of sales data to support data-driven sales decision-making.

Designed multiple types of interactive visualization components (pie charts, trend graphs, dual-axis analysis charts) in line with business-grade visual design standards, intuitively presenting truck model sales distribution, sales growth rates, customer traffic trends, and sales consultant performance.

Business Value: Replaced traditional manual Excel reporting processes, improved the efficiency of sales data processing and analysis, provided quantitative evidence

for sales strategy optimization, and enhanced operational decision-making efficiency in the global commercial vehicle market.

Streamlit:https://truck-sales-data-dashboard.streamlit.app/

Github:https://github.com/XIE-LIDONG/Truck-Sales-Data-Dashboard

Work Experience:

2025.07-2026.02 | International Sales Specialist (Commercial Vehicles) | Riyadh, Saudi Arabia

Led sales expansion of Chinese brand FAW commercial vehicles (tractor units, mixer trucks, water tankers, dump trucks, etc.) in the Saudi Arabian market, focusing on developing transportation, engineering, and logistics enterprise clients in Riyadh.

Tech-enabled customer development: Independently developed an automated data scraping tool using Google Maps**(Project-4)**, which batch-collected target enterprise information via city keywords and auto-populated Excel spreadsheets.Drove conversions through cold calling, offline showroom visits, and on-site client meetings, successfully secured a promising large-scale client prospect. (200 tractor units demand, estimated contract value of $10 million, now in advanced negotiation stage) and multiple medium-small batch orders (3-5 mixer trucks, etc.).

Built data-driven management system**(Project-7)**: Developed a visualized sales analytics dashboard for the showroom, supporting sales volume tracking, sales performance comparison, customer traffic analysis, and best-selling model ranking. Provided data-driven insights for sales strategy optimization and improved team operational efficiency.

Optimized customer retention mechanism: Addressed the pain point of customer information loss by designing a questionnaire system**(Project-6)**—offering detailed truck specification sheets as incentives for customers to provide company names and contact details. Followed up with phone calls to nurture relationships, significantly boosting customer retention rates and showroom sales.

Innovated efficiency tools: Developed an automated translator**(Project-5)** to solve cross-lingual document challenges (French/Arabic contracts and files). Integrated Google Translate API and multi-threaded processing to extract text, translate it into the target language, and replace the original content in-situ while fully preserving the document's original structure (images, tables). Reduced translation time for 40-page documents from hours to 3 minutes, greatly improving efficiency and accuracy for cross-border business communication.

2024.11-2025.06 | TikTok English Live Stream Host & Sales Manager | USA Market Focus

Led English live streaming for intangible cultural heritage (ICH) products, including Jianzhan tea sets, Jingdezhen porcelain, Longquan celadon, and Ru kiln ware, focusing on cultural value communication and premium product promotion in the

USA market.

Dual-role responsibility as host and sales manager: Achieved a monthly average sales revenue of $22,000 through high-conversion live broadcasts and strategic customer relationship management.

Managed a live streaming team: Independently developed content scripts, allocated team responsibilities, and oversaw end-to-end execution of live broadcast campaigns to ensure operational efficiency.

Conducted comprehensive market research: Analyzed competitors' pricing strategies,product positioning, and USA customer preferences to design data-driven marketing plans that enhanced product competitiveness.

Leveraged multi-channel social media (Telegram, Instagram, WhatsApp, Facebook) to build and nurture customer relationships, boosting customer loyalty and repeat purchase rates.

Handled full-cycle customer management: Addressed pre-sales inquiries, coordinated logistics and shipping, and resolved post-purchase feedback, maintaining high customer satisfaction.

Referees:

Referee Name	Position	Organization	Referee Type	Reference Letter
Zeeshan Ahmad	Ph.D,Associate Researcher	Ningbo Institute of Digital Twin, Eastern Institute of Technology	Academic	PDF-16
Qiao Mengying	Sale Manager	Hubei Zoom Yong Yuexin Automobile Industry Co., Ltd.	Professional	PDF-17